Providing a dashboard for monitoring effects of contact behavior on the spread of SARS-CoV-2

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Background: Contact Index

We feature a **unique metric**, the contact index, that is correlated to the effective reproduction number [Rüdiger et al., 2020].

It is based on **pings of anonymous mobile app users**, which corrects the raw number of contacts by weighing in secondary contacts, akin to the so-called friendship paradox.

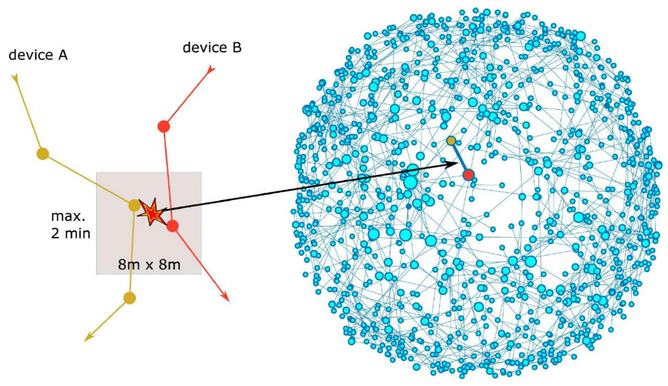

Figure 1: Visualization of contact network from pings of anonymous mobile app users. Figure taken from [Rüdiger et al., 2020].

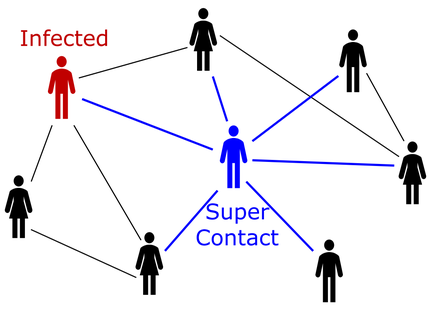

Figure 2: Visualization of importance of weighing in secondary contacts.

The correlation between the R-value and the 7-day contact index in 2020 is highest with a **lag of 20 days.**
Therefore, we display the metrics with a default shift of 20 days on the dashboard.

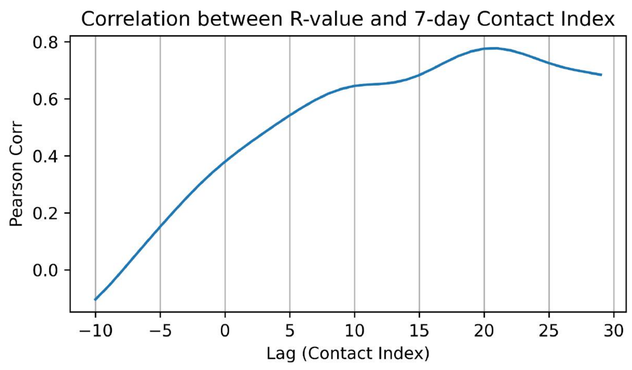

Figure 3: Correlation between R-value and 7-day contact index with different lags of the contact index.

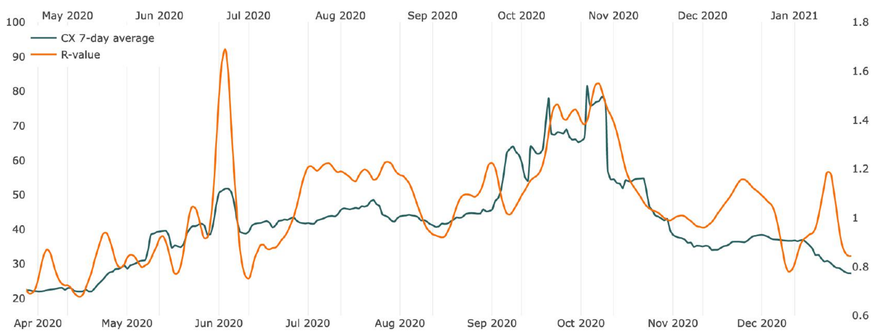

Figure 4: Graph displaying the average contact index (turquoise), shifted by 20 days, overlayed over the R-value (orange) in Germany in 2020.

The Dashboard

We have developed a **dashboard that features the impact of contacts on the spread of COVID-19** in hopes of better informing decision-makers and to increase the public's openness to contact-reducing measures, thereby decreasing case numbers.

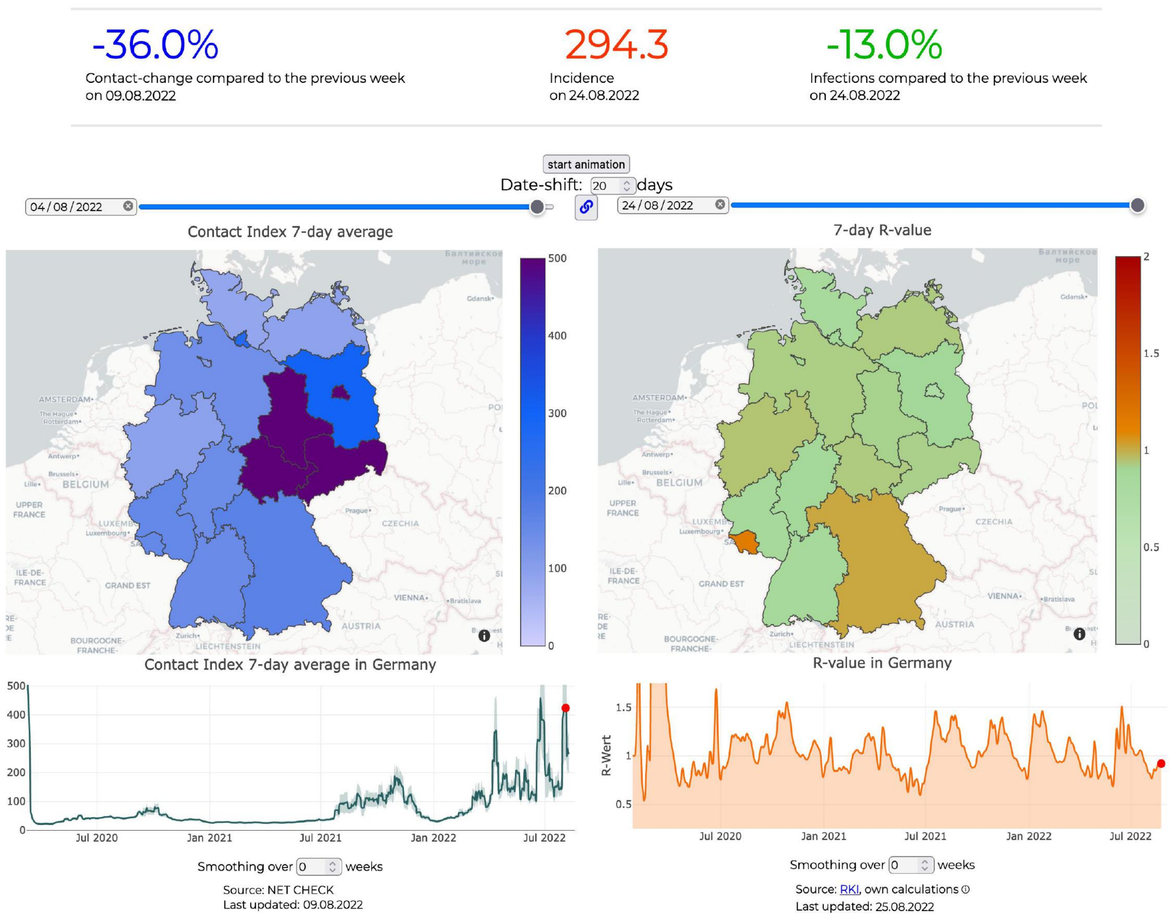

Figure 6: Dashboard showing the contact index on the left and the R-value on the right.

Key Features:

- Based on Plotly.js for **high adaptability**
- **Responsiveness**
- **Colorblind-friendly**
- Explanations for **general audience**

QR code for our dashboard: https://contactindex.netcheck.de

Effective R-Value

- **Double smoothing** to counter reporting delays in infections in Germany

$$s_n = \frac{\sum_{i=n}^{n+6} \frac{\text{new infections}}{4}}{7}$$

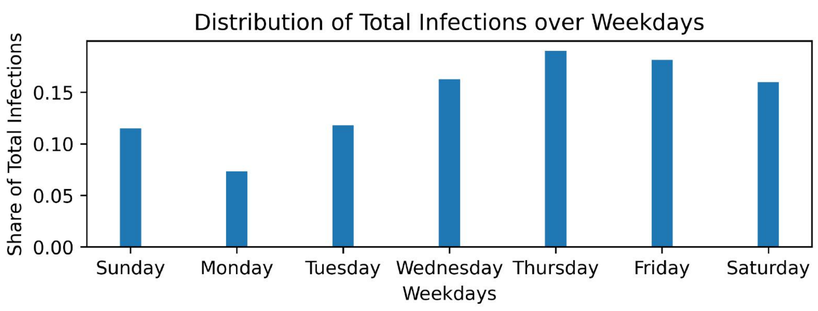

Figure 5: Distribution of infections over weekdays in Germany in 2020.

- While assuming an **incubation time of 4 days,** we calculate the R-value from the smoothed infection numbers

$$R_n = \frac{s_n}{s_{n-4}}$$

Infection Spread

We also feature **infection data spanning the whole length of the pandemic**, with many **interactive tools** in order to create a comprehensible overview of the progression of SARS-CoV-2.

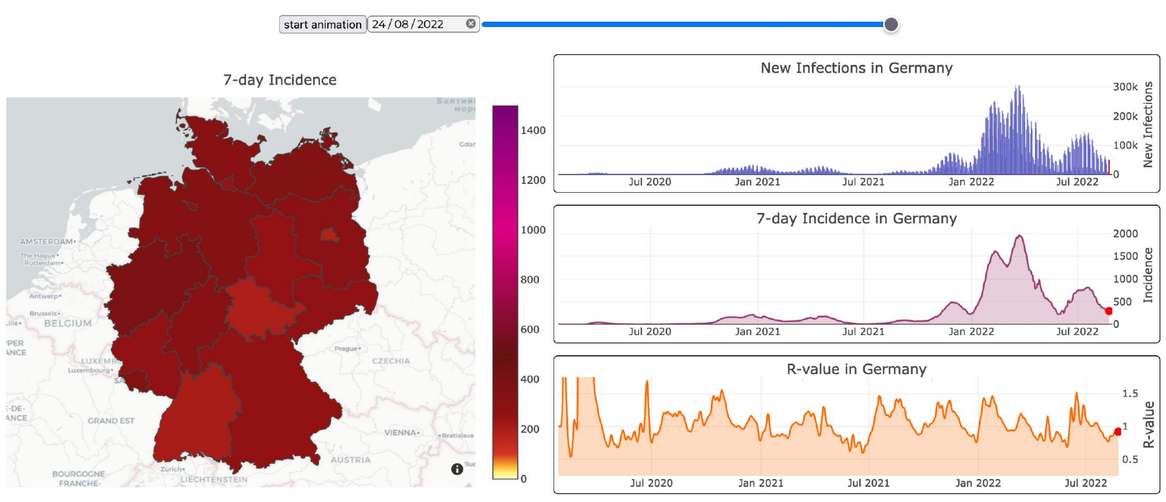

Figure 7: Dashboard showing the 7-day incidence of the selected day on the left and graphs of the new infection numbers, 7-day incidence and the R-value over the whole pandemic on the right.

Interactive Tools:

- Select **each state** to see their course of the pandemic
- Watch an **animation** of the spread throughout Germany
- **Zoom in** on the graphs to better see a specific timeframe

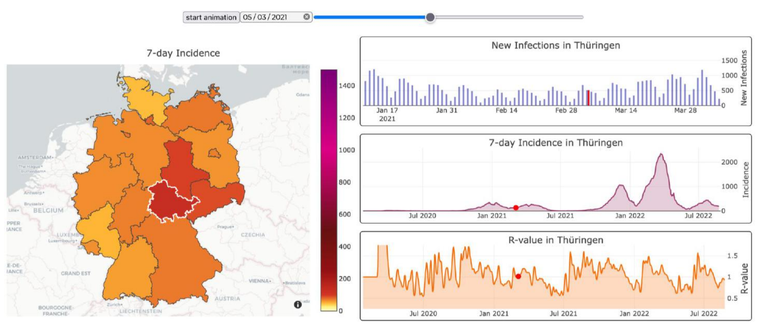

Figure 8: Dashboard showcasing the infection spread of a selected state.

Reference: Rüdiger, Sten, et al. "Predicting the SARS-CoV-2 effective reproduction number using bulk contact data from mobile phones." Proceedings of the National Academy of Sciences 118.31 (2021): e2026731118

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